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Algal Organic Matter Drives Methanogen-Mediated Methylmercury Production in Water from Eutrophic Shallow Lakes

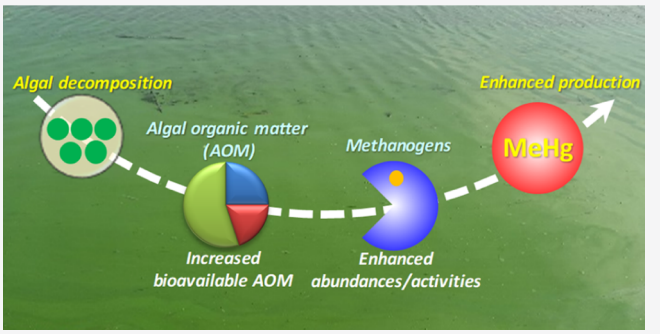
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ABSTRACT: Algal blooms bring massive amounts of algal organic matter (AOM) into eutrophic lakes, which influences microbial methylmercury (MeHg) production. However, because of the complexity of AOM and its dynamic changes during algal decomposition, the relationship between AOM and microbial Hg methylators remains poorly understood, which hinders predicting MeHg production and its bioaccumulation in eutrophic shallow lakes. To address that, we explored the impacts of AOM on microbial Hg methylators and MeHg production by characterizing dissolved organic matter with Fourier transform ion cyclotron resonance mass spectrometry (FTICR-MS) and three-dimensional excitation-emission matrix (3D-EEM) fluorescence spectroscopy and quantifying the microbial Hg methylation gene *hgcA*. We first reveal that the predominance of methanogens, facilitated by eutrophication-induced carbon input, could drive MeHg production in lake water. Specifically, bioavailable components of AOM (*i.e.*, CHONs such as aromatic proteins and soluble microbial byproduct-like materials) increased the abundances (Archaea-*hgcA* gene: 438−2240% higher) and activities (net CH_4 production: 16.0−44.4% higher) of Archaea (*e.g.*, methanogens). These in turn led to enhanced dissolved MeHg levels (24.3−15,918% higher) for three major eutrophic shallow lakes in China. Nevertheless, our model results indicate that AOM-facilitated MeHg production could be offset by AOM-induced MeHg biodilution under eutrophication. Our study would help reduce uncertainties in predicting MeHg production, providing a basis for mitigating the MeHg risk in eutrophic lakes.



KEYWORDS: *methylmercury, organic matter, methanogens, eutrophication, lake*

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1. INTRODUCTION

Methylmercury (MeHg) biomagnification through food chains and thus dietary exposure risks raise global concerns.[1−3] For instance, it was estimated that protecting against the neurodevelopmental toxicity of MeHg in children by reducing environmental MeHg exposure in the European Union could save more than 8 billion euros per year.[4] The risk of MeHg in aquatic systems is primarily governed by the microbial Hg methylation process.[5] It is well documented that Hg methylation mainly occurs in sediments.[6,7] Nevertheless, recent studies provided initial evidences showing that methylation of Hg could be mediated by various environmental communities in lake water, including periphyton,[8] biofilms,[9] algae,[10] or planktonic detritus.[11] The produced MeHg in lake water is more accessible to primary benthic consumers than that in sediments, which can be subsequently accumulated in fish *via* trophic transfer, posing health risks.[12] Therefore, the process of Hg methylation in lake water, as well as its controlling factors, warrants further investigation.

Algal organic matter (AOM) has been reported to play a critical role in controlling microbial Hg methylation (*e.g.*, by fueling the metabolism of microbial methylators).[13,14] The global occurrences of algal blooms have expanded during past decades,[15,16] and some dominating cyanobacteria release high amounts of AOM, a major fraction of dissolved organic matter (DOM), into water, especially in shallow lakes.[17,18] However, the complexity of AOM compositions[19] and its dynamic changes in eutrophic shallow lakes[20] make it difficult to understand microbial Hg methylation in water. For example, during algal decomposition, AOM was depleted in highly unsaturated, high-oxygen molecules and enriched in aliphatic, highly unsaturated, low-oxygen molecules, and these dynamic changes in AOM compositions in turn impacted microbial diversity[21] and could possibly regulate microbial Hg methylation in water. On the other side, various microbial Hg methylators, such as sulfate-

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reducing bacteria (SRB), methanogenic Archaea, iron-reducing bacteria (FeRB), and other syntrophs,[22] may have diverse responses to carbon input.[23] While microbial Hg methylators' abundances are rarely reported in lake water,[24,25] their roles in Hg methylation in response to AOM input are also unclear. Consequently, the complexity in both AOM compositions and the diversities of microbial methylators make it challenging to predict MeHg risk, particularly in eutrophic shallow lakes.

Recently, ultrahigh-resolution Fourier transform ion cyclotron resonance mass spectrometry (FTICR-MS) has been used to determine thousands of individual molecular formulae within AOM samples.[14,20,21] Coupled with three-dimensional excitation-emission matrix (3D-EEM) fluorescence spectroscopy, this technique can facilitate the identification of potential factor controls of different AOM components on various microbes.[21] Meanwhile, the recently identified Hg methylation gene *hgcA* and the development of primers could help quantify various microbial methylators (*e.g.*, Archaea and Deltaproteobacteria).[5,26,27] Therefore, these techniques could reveal microbe dynamics and their responses to AOM input in eutrophic lakes.

To explore the potential impacts of AOM on microbial MeHg production in lake water, we conducted a series of microcosm experiments as well as an *in situ* incubation using water and algae sampled from three major eutrophic shallow lakes in China. The underlying mechanisms were explored by (1) characterizing AOM components (*i.e.*, FTICR-MS and 3D-EEM), (2) quantifying the abundances of various microbial Hg methylators (*e.g.*, Archae- and Deltaproteobacteria-*hgcA* gene abundance) at different stages of algal decomposition, and (3) elucidating the responses of various microbial Hg methylators and MeHg production to AOM input. Furthermore, to predict MeHg accumulation in fish in eutrophic lakes, we established a eutrophic lake MeHg risk model, by incorporating the AOM-MeHg production relationship quantified in this study. This work provides insights toward understanding the Hg biogeochemistry in eutrophic lakes and helps improve the prediction of MeHg risk in aquatic systems.

2. MATERIALS AND METHODS

2.1. Collection of Water and Algal Samples from Eutrophic Lakes. The water and algal samples were collected from three eutrophic shallow lakes across a wide longitude range of China and were used in the microcosm experiments as described below. Eutrophication is favored in shallow lakes,[18] which is particularly prominent in China.[28] These three representative lakes, include Chaohu Lake (CHL, the fifth largest freshwater lake in China), Donghu Lake (DHL, the second-largest urban lake in China), and Dianchi Lake (DCL, the largest freshwater lake in the subtropical plateau of Southwest China). The average water depths of the three studied lakes were 4.09 m for CHL, 2.50 m for DHL, and 4.40 m for DCL (Supporting Information, Table S1). All of these lakes are typical shallow lakes and they all serve as sources of drinking water. However, eutrophication has been a major ecological problem in these lakes due to anthropogenic activities from surrounding cities over the past decades.[29] During algal blooms in mid-August in 2019, samples of algal biomass and the corresponding lake water used in the microcosm experiments (described in Section 2.2) were collected at the algae-gathering areas in the three lakes (Figure S1).

Before collecting samples, all devices and tools for Hg measurements were acid-washed and rinsed with ultrapure water and lake water. The algal samples were collected using a plankton net (112 μm mesh-size). Surface lake water samples were collected from a nearby area (without gathering algae). The algae and water samples were put into a cooler with ice packs and immediately transported to the laboratory. Algal samples were rinsed, freeze-dried, ground, and then stored at −20 °C until further analysis. Dissolved water samples were preserved with 0.1% HCl for determination of MeHg and total Hg (THg) levels (Text S1), and the unfiltered water samples were used for microcosm experiments.

2.2. Microcosm Experiments. Three microcosm experiments were designed to elucidate the effects and mechanisms of AOM input on microbial Hg methylation in eutrophic lakes. Details of all treatments are summarized in Table S2.

Experiment A was conducted to explore the effects of AOM input on microbial Hg methylation by comparing the concentrations of dissolved MeHg in water with ("Algae") or without algal biomass ("Control"). This experiment has two subsections: A-1, quantifying the production of MeHg in response to AOM from different lakes; A-2, adding different amounts of algal biomass from CHL to investigate AOM-impacted MeHg production under different scenarios of eutrophication (*e.g.*, from oligotrophication to severe eutrophication). For A-1, 40 mL unfiltered water was put into a 50 mL tube (Corning, USA). Mercury (*i.e.*, $HgCl_2$) was added in our microcosm experiments, simulating exogenous Hg input into lake systems, for example, *via* runoff or atmospheric deposition.[30,31] The spiked Hg level (equivalent to 500 ng·L^{-1}) is within the range of Hg concentrations reported in three studied lakes (*e.g.*, 50−1050 ng·L^{-1}[32,33]) and comparable with that reported in urban storm water runoff (*e.g.*, 173 to 8254 ng·L^{-1}, Nanjing City, Eastern China[34]). It has been widely recognized that ionic Hg would quickly complex with DOM, especially in high dissolved organic matter (DOC) levels, leading to most of spiked Hg as Hg−DOM complexes.[35] From each "Algae" treatment, 0.01 g of dried algal biomass was added to the tubes. This resulted in about 2000 μg·L^{-1} chlorophyll a (Chl a) level in the mixtures (*i.e.*, water and algal biomass), which was comparable to field observations in eutrophic lakes during algal blooms, for example, 2000−3000 μg·L^{-1} at the algae-gathering area.[36] Because the amount of MeHg in the added algal biomass (pg) contributed to <10% of MeHg in water, the contribution of algae-associated MeHg was considered negligible for the "Algae" treatments. All tubes were tightly capped and sealed with Parafilm, and then the mixtures were blended using a vortex mixer for 1 min. Three replicates were made in the "Algae" treatment (lake water amended with algal biomass) or "Control" group (lake water without algal biomass added). To mimic the water environment during the period of algal decomposition in eutrophic shallow lakes, the tubes were kept in the dark (to avoid MeHg photodegradation) at 23 °C (*i.e.*, average temperature during algal decomposition in Chaohu Lake Basin). The water in shallow lakes is maintained in dynamic contact due to active water mixing, and stratification is generally not observed in shallow lakes.[18] Therefore, our microcosm experiments were conducted with mixing, specifically, the tubes were shaken twice per day for homogeneity during the whole incubations. At each incubation time point, that is, days 3, 12, and 24, the mixtures were centrifuged at 4000*g* for 20 min, and the supernatants were filtered. The filtrates were determined for the concentrations of dissolved MeHg to indicate MeHg production. For A-2, Hg-spiked lake water was then added with different amounts of algal biomass (taking a sample of Chaohu Lake's algae) (Figure S2),

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that is, 0 ("Control"), 0.001 g ("200 Algae"), 0.005 g ("1000 Algae"), 0.01 g ("2000 Algae"), and 0.03 g ("6000 Algae") to simulate different nutritional statuses of lakes. The levels of Chl a were 15.3 ± 1.76 $\mu g \cdot L^{-1}$ for "Control" (oligotrophication), 241 ± 19.4 for "200 Algae" (mild eutrophication), 969 ± 20.0 $\mu g \cdot L^{-1}$ for "1000 Algae" (moderate eutrophication), 1919 ± 59.0 $\mu g \cdot L^{-1}$ for "2000 Algae", and 6007 ± 93.5 $\mu g \cdot L^{-1}$ for "6000 Algae" (severe eutrophication). All incubation procedures and conditions were the same as those in A-1. Before sampling on "day 0", the mixed solution was incubated for 4 h to pre-equilibrate a Hg(II) spike in a DOC-contained lake water. The sampling time was set as day 0 (*i.e.*, 4 h), 3, 7, 12, 18, and 24, to explore the dynamic characteristics of MeHg production for all treatments.

Experiment B was carried out to elucidate the underlying mechanisms of AOM-impacted MeHg production, which was achieved by determining AOM composition and bioavailability, the abundances and activities of various microbial Hg methylators, and further exploring their relationships in response to AOM input. This experiment was similar to the "2000 Algae" treatment (*i.e.*, 0.01 g of Chaohu Lake's algal biomass addition) in experiment A-2 mentioned above. In this experiment, the pre-punched tubes were sealed with silicone rubber. After sealing, the tubes were shaken for 1 min to equilibrate CH_4 between the water and headspace. At each sampling time point, 5 mL samples of gas was taken from the headspace using a syringe needle to determine the production of CH_4 (indicating the activities of methanogens[37]). Then, the mixtures were centrifuged at 4000*g* for 20 min and the supernatants were filtered. The level of DOC was measured using a total organic carbon analyzer (TOC-VCPH, Shimazu). Dynamic changes of DOC concentrations with increasing time (t) during the decomposition of DOM was fitted by a G model ($DOC = C_1e^{-k_1t} + C_2e^{-k_2t} + C_3e^0$), which consisted of the sum of exponential decay terms for three various bioavailability fractions: labile (C_1), semi-labile (C_2) and recalcitrant DOC (C_3).[21] The spectral characteristics of DOM were acquired by a HITACHI F-7000 spectrofluorometer. The components of DOM, that is, aromatic protein I (API), aromatic protein II (APII), fulvic acid-like, soluble microbial byproduct-like (SMP), and humic acid-like materials were quantified by the fluorescence region integration method[13,38] (Text S2). DOM in the filtrates from days 0 (*i.e.*, 4 h), 3, and 24 was extracted with the solid phase extraction method and then molecularly characterized by using a Fourier transform ion cyclotron resonance mass spectrometer (Bruker SolariX 15.0 T) equipped with an electrospray ionization (ESI) source.[21,39] The methods of isolation, preconcentration, desalination, and analysis of DOM compositions are described in Text S3. The assigned formulas were categorized based on the modified aromaticity index (AI_{mod}) and H/C or O/C ratio. The double bond equivalence (DBE) and AI_{mod} were calculated to indicate the dynamic changes in aromaticity and oxidation state of DOM; for example, a lower value of AI_{mod} was found in labile and semilabile fractions than that of refractory molecules, indicating the negative relationship between aromaticity and bioavailability of DOM.[21] The molecular lability (MLB_L) was used to estimate the lability of DOM, and a higher MLB_L meant DOM with a higher lability.[20] Aliquots of filtered water were acidified with trace metal grade HNO_3 and then analyzed for the changes in dissolved Fe levels between the "Control" groups and "Algae" treatments (indicating iron reduction[40,41]) by ICP−MS.[42] Other aliquots were analyzed for the changes in dissolved sulfate levels (indicating sulfate reduction[43]) using a Dionex ICS1100 (Thermo Fisher Scientific, USA). The levels of MeHg in water were determined. The abundance of the *hgcA* gene in the mixture was quantified using the clade-specific degenerate primer pairs ORNL-Delta-HgcA and ORNL-Archaea-HgcA for Deltaproteobacterial methylators (*e.g.*, SRB or FeRB) and Archaeal methylators (*e.g.*, methanogens),[27,44] respectively (Text S4).

Experiment C was conducted to further explore AOM-impacted microbial MeHg production mechanisms by identifying the Hg methylation rate (k_m) of the key microbial Hg methylators under AOM input. This was achieved by quantifying k_m using Hg-stable isotope tracers in the presence of specific inhibitors for different microbial methylators (*i.e.*, SRB and methanogens). The unfiltered lake water, amended with algal biomass (the added amount set as the same as that used in experiment A-1) and specific inhibitors simultaneously, was incubated for 12 and 24 days. For SRB, sodium molybdate (Mo) within 20 mM was used as the specific inhibitor, as commonly used in previous studies.[45,46] For methanogens, a higher amount (100 mM) of 2-bromoethanesulfonate acid (BESA) was used as the specific inhibitor, considering the generally high abundance of methanogens in periphyton or flocculent materials in lake systems.[47,48] The basis for the used amount was given by a preliminary experiment that added different amounts of specific inhibitors (detailed in Text S5). A total of four treatments were set as follows: (1) "Control": unfiltered lake water (40 mL) without any algal biomass or inhibitor addition; (2) "Algae": unfiltered lake water (40 mL) amended with 0.01 g of algal biomass but with no inhibitor; (3) "Mo": the same as "Algae" but with 20 mM Mo amendment; and (4) "BESA": the same as "Algae" but with 100 mM BESA amendment. At each sampling point (*i.e.*, days 12 and 24), all treatment mixtures were spiked with $^{202}HgCl_2$ (Oak Ridge National Laboratory, TN, USA) at concentrations similar to ambient values. Within 5 min after isotope addition, three triplicates (samples of "t_0") were frozen at −80 °C to determine k_m further. Other triplicates were centrifuged (4000*g* for 10 min) and filtered for analysis of *hgcA*-gene abundance. For each treatment, three replicates spiked with the Hg stable isotope (*i.e.*, $^{202}HgCl_2$) were incubated for 24 h as the samples of "t_f" at 23 °C. A similar sampling protocol as for t_0 was used for microbial analyses. The determination and calculation of the methylation rate (k_m) with AOM and inhibitor addition are presented in Text S6. The detailed results and discussions of SRB and methanogens' relative contributions to microbial Hg methylation are presented in Text S7.

Additionally, in order to scale up the results from lab microcosms to the field process, a 3 day incubation study was conducted *in situ* to investigate the effect of AOM on Hg methylation. The detailed procedures and results of this *in situ* incubation are shown in Text S8.

2.3. Establishment of a Eutrophic Lake MeHg Risk Model. To predict changes in MeHg exposure risk in response to eutrophication-driven MeHg production in water, a eutrophic lake MeHg risk model was established by considering AOM-facilitated MeHg production (quantified in this study) as well as AOM-impacted MeHg bioaccumulation and AOM-impacted MeHg photodegradation (based on the literature). The detailed steps of model establishment and output results are listed in Text S9, which are briefly shown as follows: (1) levels of MeHg in lake water under different eutrophication scenarios (taking Chaohu Lake as an example) were estimated. This is achieved by

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obtaining values of Hg methylation potential (*i.e.*, MeHg %, quantified by experiment A-2 in this study), THg levels in lake water (collected from previous reports[34,49]), and incorporating DOC-impacted MeHg photodegradation.[50] (2) The probability density and cumulative probability of MeHg in fish were acquired by considering DOC-adjusted bio-concentration factors (BCFs) of MeHg under different eutrophication scenarios.[51] (3) The human exposure risk of MeHg from fish consumption was assessed by the points of per-fetus IQ decrements.[52,53] Monte Carlo Simulation (MCS) was used to quantitatively estimate the uncertainty. Our estimated median MeHg levels in fish and human exposure risk were comparable to those reported values in previous studies,[34,53] showing the reliability of the results.

2.4. Statistical Analysis. Statistical analyses and MCS used in a eutrophic lake MeHg risk model were performed with *R* software (win10 *R*-4.0.3). One-way analysis of variance (ANOVA) with Tukey's multiple comparison tests was applied to compare the statistical significance between the "Control" and "Algae" treatments. The lm() function was used for linear regressions.

3. RESULTS AND DISCUSSION

3.1. AOM Enhanced MeHg Production in Water from Eutrophic Shallow Lakes. By examining different lakes and eutrophication scenarios in microcosm experiments, we demonstrate that AOM input could enhance MeHg production in water from eutrophic shallow lakes. For the examined lakes, algal addition resulted in 975−10,834% (Chaohu Lake, CHL), 2336−10,331% (Donghu Lake, DHL), and 91−366% (Dianchi Lake, DCL) increases in MeHg levels (compared with "Control" without algal addition, $p < 0.05$, Figure 1A) of lake water collected from three major eutrophic shallow lakes in China. The relative ratio of MeHg levels (*i.e.*, "Algae"/"Control") showed a downward trend in CHL and DHL at the later stage of incubation (*e.g.*, days 12−24, Figure S3), which could be due to the consumption of bioavailable DOM. It was partly supported by the decreased abundances of microbial Hg methylators (discussed in Section 3.3). In contrast, this ratio in DCL showed an increasing trend. This inconsistency may be related to the differences in DOM compositions[54] and microbial communities among different lakes. Under different eutrophication scenarios (for CHL only), significant increases in MeHg production ("Algae" *vs* "Control", $p < 0.05$) were observed in the "Algae" treatments (except "200 Algae"). Specifically, "1000 Algae", "2000 Algae", and "6000 Algae" AOM addition resulted in 24.3−645% (average 207%), 1372−13,904% (average 6427%), and 331−15,918% (average 368%) increases in MeHg levels, respectively (Figure 1B). Results of *in situ* incubation also indicated that AOM input could promote MeHg production in eutrophic lake water (details described in Text S8). The enhanced MeHg production, in response to AOM input, was further supported by the increased Hg methylation rate (k_m): AOM addition significantly increased the values of k_m (2.2 and 152 times higher on days 12 and 24, respectively) in lake water compared with "Control" (Figure S4). A relatively low k_m value was observed in the "Control" group on both days 12 and 24, which may be due to the low Hg bioavailability in this group. For instance, DOM at relatively low levels [low DOM:Hg(II) ratios] facilitates the reduction of Hg(II) and thus reduces Hg bioavailability under anoxic conditions.[35]

Although AOM could significantly increase MeHg production in water (*i.e.*, 1−2 orders of magnitude higher), such effects on fish MeHg levels could be offset by AOM-induced MeHg biodilution. To predict changes in MeHg exposure risk in response to eutrophication-driven MeHg production in water, a eutrophic lake MeHg risk model was established. With multiple effects of AOM on MeHg bioaccumulation involved (details described in Text S9), we demonstrate that (1) mild and moderate eutrophication (Chl a level in water: 200−1000 $\mu g \cdot L^{-1}$, Table S2) could reduce the accumulation of MeHg in fish. Specifically, the Monte Carlo Simulation showed that the estimated median MeHg levels in fish under mild eutrophication (*i.e.*, "200 Algae", 710 $ng \cdot g^{-1}$) and moderate eutrophication (*i.e.*, "1000 Algae", 618 $ng \cdot g^{-1}$) were 12.2 and 8.0% lower than that of the "Control" (772 $ng \cdot g^{-1}$), respectively. The decreased MeHg bioaccumulation under mild and moderate eutrophication could be related to the impact of eutrophication-induced biodilution of MeHg, that is, increasing algal biomass reduces Hg accumulation at higher trophic levels through the dilution of Hg in the consumed biomass.[55,56] This lowered the values of $BCF_{water-seston}$, leading to lower MeHg in consumers at higher trophic levels.[51] (2) In contrast, severe eutrophication (Chl a level in the water: approximately 2000 $\mu g \cdot L^{-1}$, Table S2) could enhance MeHg accumulation in fish from eutrophic lakes. Specifically, the simulation showed that the estimated median MeHg levels in fish of "2000 Algae" (862 $ng \cdot g^{-1}$) were 11.7% higher than those of "Control" (Text S9). This could be mainly due to the promotion of Hg methylation induced by AOM and the AOM-impacted inhibition of MeHg photodegradation.[50]

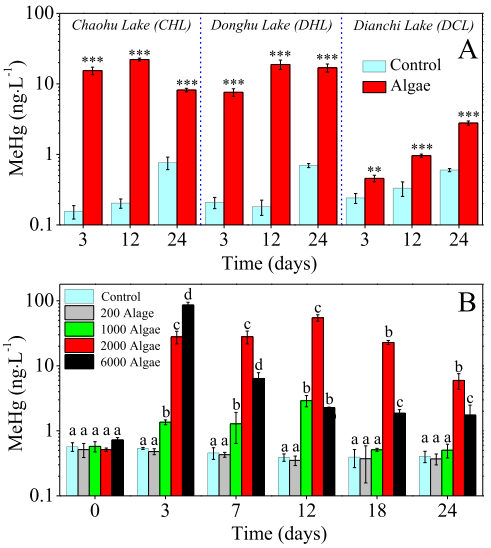


Figure 1. Levels of MeHg in water during 24 days of incubation. Three eutrophic lakes, that is, Chaohu Lake (CHL), Donghu Lake (DHL), and Dianchi Lake (DCL), were selected in experiment A-1 (A). Different amounts of algal biomass from CHL were added in experiment A-2 (B). Error bars show standard errors of the mean, $n = 3$. The asterisk above the bars in Figure 1A indicates a significant difference between the "Control" group and the "Algae" treatment (one-way ANOVA, *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; without *, $p > 0.05$). Different letters (a−d) above the bars in Figure 1B denote significant differences among different treatments (one-way ANOVA, $p < 0.05$). "200 Algae", "1000 Algae", "2000 Algae", and "6000 Algae": unfiltered lake water (40 mL) amended with 0.001, 0.005, 0.01, and 0.03 g of algal biomass from CHL in experiment A-2, respectively. "Control": the same unfiltered lake water with no algal addition. Before sampling on "day 0", the mixed solution was incubated for 4 h to pre-equilibrate a Hg(II) spike in DOC-contained lake water.

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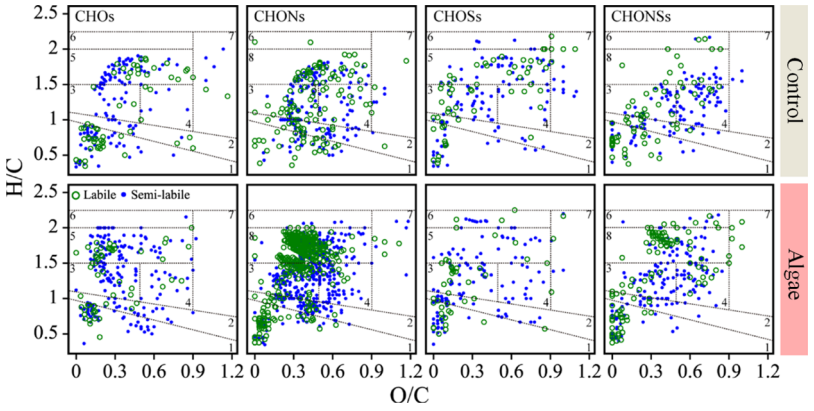


Figure 2. Distribution and composition of the bioavailable components (*i.e.*, labile and semilabile fractions) of DOM in the "Control" (upper panels) groups and "Algae" treatments (lower panels). Van Krevelen diagram is applied to visually show the composition of DOM by plotting with O/C as *x* axes and H/C as *y* axes. Eight typically functional groups were defined based on the modified aromaticity index (AI_{mod}) and the H/C or O/C ratio: (1) condensed aromatics ($AI_{mod} > 0.67$), (2) aromatics ($0.5 < AI_{mod} \leq 0.67$), (3) highly unsaturated, low oxygen ($AI_{mod} < 0.5$, O/C < 0.5, and H/C < 1.5), (4) highly unsaturated, high oxygen ($AI_{mod} < 0.5$, $0.5 \leq$ O/C < 0.9, and H/C < 1.5), (5) aliphatics (N = 0, O/C < 0.9, and $1.5 \leq$ H/C ≤ 2), (6) saturated fatty acids (O/C < 0.9 and H/C > 2), (7) sugars (O/C $\geq$ 0.9), and (8) peptide-like compounds (N > 0, O/C < 0.9, and $1.5 \leq$ H/C ≤ 2).[21] The AI_{mod} DBE and the molecular lability index (MLB_L) were calculated to indicate the dynamic changes in the bioavailability of DOM: $AI_{mod} = (1 + C - 0.5O - S - 0.5H)/(C - 0.5O - S - N - P)$; $DBE = 1 + C - 0.5H + 0.5N + 0.5P$; the MLB_L was calculated by using the total intensity of molecular formulae with H/C $\geq$ 1.5 to the total intensity of molecular formulae.[20] Lower values of AI_{mod}, DEB, and higher MLB_L were found in labile and semilabile fractions than those of refractory molecules, indicating a higher lability of DOM.[20,21]

Both these factors could contribute to higher MeHg accumulation in eutrophic water.

Findings from the present study suggest that the dual effects of AOM on MeHg bioaccumulation could explain the inconsistent relationships between eutrophication degree and MeHg bioaccumulation observed in previous studies. For instance, according to a national scale investigation of fish MeHg in Korea, MeHg concentrations in fish showed a negative correlation with eutrophication levels (indicated by Chl a: r^2 = 0.90; total phosphorus: r^2 = 0.75).[56] However, under hypereutrophic conditions, bioconcentration rather than biodilution was observed to govern MeHg bioaccumulation in zooplankton in Onondaga Lake over 30 years.[57] Interpreting those observed inconsistencies caused by the dual effects of AOM on MeHg bioaccumulation would help better understand MeHg risk in eutrophic lakes.

3.2. Algal Decomposition Brought Bioavailable Fractions of DOM. The mechanisms of AOM-induced MeHg production could be explained as follows: first, algal decomposition increased the amount of bioavailable fractions of DOM (this section); second, bioavailable components of AOM increased the abundances/activities of microbial Hg methylators, particularly Archaea (*e.g.*, methanogens, discussed in Section 3.3); and third, methanogens may drive MeHg production in lake water during algal decomposition (discussed in Section 3.4).

Increase in the levels of both the total amount and the specific components of DOM in water were observed ("Algae" *vs* "Control", detailed in Text S2, Figures S5 and S6). Particularly, algal decomposition increased bioavailable DOM in water based on DOM degradation model results, atomic ratio compositions, and aromaticity/lability index quantified by FTICR-MS. First, higher bioavailability of AOM (*i.e.*, the sum of labile and semilabile DOC, according to G model fitting, Figure S5) was found in "Algae" than "Control" (69% *vs* 39%). Second, higher percentages of CHONs, a major bioavailable fraction of DOM in "Algae" (64.5 and 46.1% of labile and semilabile fractions, respectively, Table S3), were observed under AOM input (46.0 *vs* 40%, Figure 2, Table S4). The high bioavailability of CHONs might arise from their higher H/C ratios and saturation.[21] Third, a higher bioavailability of DOM in "Algae" than in "Control" could be further evidenced by the lower values of AI_{mod} (modified aromaticity index, 0.16) and DBE (6.69) and higher value of MBL_L (molecular lability index, 0.46 on day 0) (Table S4), all of which have been reported to be associated with DOM bioavailability.[21]

3.3. Bioavailable Components of AOM Increased the Abundances of Archaea in Water from Eutrophic Shallow Lakes. Our results show that AOM input could increase the abundances of different microbial Hg methylators. The input of AOM resulted in 438−2240 and 52.1−38,844% increase in the copy numbers of Archaea-*hgcA* gene and Deltaproteobacteria-*hgcA* gene, respectively ("Algae" *vs* "Control", $p < 0.05$, Figure 3). Specifically, ΔArchaea-*hgcA* ("Algae"

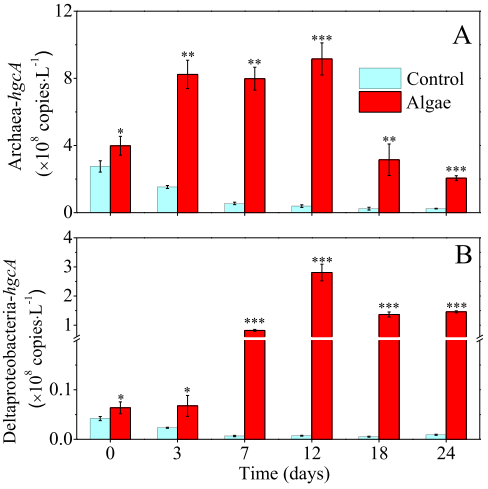


Figure 3. Abundance of Archaea- (A) and Deltaproteobacteria-*hgcA* (B) gene during 24 days of incubation. "Algae": unfiltered lake water (40 mL) amended with 0.01 g of algal biomass from CHL. "Control": the same unfiltered lake water with no algal addition.

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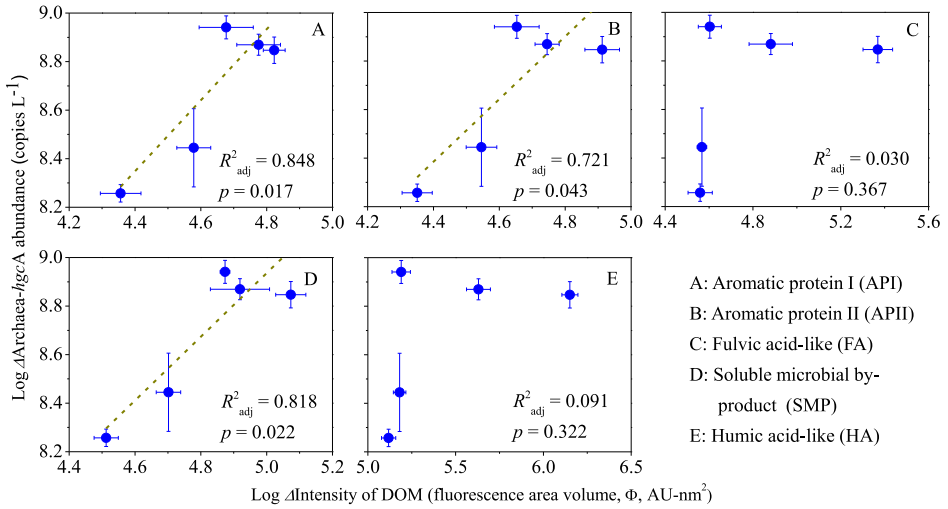


Figure 4. Relationships between the incremental Archaea-*hgcA* gene abundance ("Algae" minus "Control", *i.e.*, ΔArchaea-*hgcA*) and the incremental intensity of five DOM components (Δintensity of DOM). Data of day 0 (*i.e.*, 4 h) were excluded when fitting linear lines due to the limited time for microbial methylators' responses.

minus "Control") abundance (2.01×10^8 to 9.15×10^8 copies·L^{-1}), in response to AOM input, was 1−2 orders of magnitude higher than ΔDeltaproteobacteria-*hgcA* abundance (0.07×10^8 to 2.81×10^8 copies·L^{-1}) (Figure S7A). Particularly during the first 12 days, more pronounced increases in *hgcA* abundance were observed for Archaea (8.23×10^8 to 9.15×10^8 copies·L^{-1}) than Deltaproteobacteria. Then, ΔArchaea-*hgcA* abundance decreased with time, which could be attributed to the exhaustion of labile carbon sources (Figure S7B). Thus, we speculate that Archaea and Deltaproteobacteria may have different responses to AOM addition, which could be explained by their competition for bioavailable DOM fractions, as discussed in the following paragraph.

For *hgcA*-containing Archaea (*e.g.*, methanogens[27]), the sharp increase in abundance could be attributed to the increased bioavailable fractions of AOM. This could be supported by the significant positive relationships between ΔArchaea-*hgcA* copies and the Δintensity of specific DOM components, including aromatic protein I, II (API, APII), and SMP:[38,58] R_{adj}^2 = 0.848, 0.721, or 0.818, respectively ($p < 0.05$, Figure 4A,B,D). Recent studies found that AOM components with less aromatic, more oxygenated, and nitrogen-rich molecules in low molecular weights (*e.g.*, <0.5 kDa) could serve as labile carbons for microbial methylators.[13,21,59] These components mainly referred to API, APII (*e.g.*, tyrosine), and SMP (*e.g.*, tryptophan phenylalanine or tryptophan) in this work. For *hgcA*-containing Deltaproteobacteria (*e.g.*, SRB or FeRB[27]), the moderate increases (compared to Archaea) in abundance could be attributed to the competition for labile carbon between Archaea and Deltaproteobacteria (Figure S7A). Different from Archaea, negative linear relationships were noted between ΔDeltaproteobacteria-*hgcA* abundance and Δintensity of bioavailable DOM fractions (*i.e.*, API, APII, or SMP, R_{adj}^2 = 0.716, 0.763, or 0.839, $p < 0.05$, Figure S8A,B,D). This might be because the initial abundance of Deltaproteobacteria was 1−2 orders of magnitude lower than that of Archaea (Figure 3); thus, Archaea could compete over Deltaproteobacteria for carbon sources. Similarly, competition between methanogens and SRB for available substrates or electron donors has been well documented.[23,37] The competition between methanogens and SRB depend on the substrates, the specific metabolic types, and their initial relative abundances.[60,61] For example, aceticlastic methanogens have been reported to prevail in the presence of sulfate.[62] Besides, high levels of low-molecular-weight substances (*e.g.*, thiols or acetic acid) could be produced during algal decomposition,[63,64] which could facilitate the growth of some specific methanogens (*e.g.*, aceticlastic methanogens). The different responses of Archaea and Deltaproteobacteria to AOM might thus lead to their different contributions to MeHg production in lake water during algal decomposition, as discussed in the next section.

3.4. Methanogen-Driven MeHg Production in Lake Water with AOM Input during Algal Decomposition. We further demonstrate the dominant role of Archaea (*e.g.*, methanogens) in driving MeHg production in water from eutrophic shallow lakes, which is supported by the following findings:

(1) The growth of methanogens, facilitated by AOM (discussed above), could in turn drive MeHg production. A positive relationship between archaea-gene abundance and MeHg levels within AOM input was observed (R_{adj}^2 = 0.832, $p < 0.05$, Figure S9A). In contrast, there was no relationship between Deltaproteobacteria-gene abundance and MeHg levels in the "Algae" treatments ($p > 0.05$, Figure S9B). These results may indicate that the enhanced production of MeHg in eutrophic lakes was largely due to the increased abundance of archaea (*e.g.*, methanogens) but not Deltaproteobacteria (*e.g.*, SRB or FeRB).

(2) The activities of methanogens were significantly enhanced by AOM during algal decomposition, contributing to increased MeHg production. Our results showed that the net CH_4 production (indicating activities of methanogens[37]) was 16.0−44.4% (average 30.5%) higher following algal biomass addition (Figure S10C). In contrast, the levels of both dissolved sulfate (indicating the activities of SRB[43]) and Fe (indicating the activities of FeRB[40,41]) showed no difference between the "Control" and "Algae" treatments ($p > 0.05$, Figure S10A,B). These results probably indicated that the activities of methanogens but not SRB and FeRB were enhanced by AOM input. Furthermore, a linear relationship was found

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between CH_4 production and MeHg levels only (R_{adj}^2 = 0.718, $p < 0.05$, Figure S11) but not between sulfate or iron reduction, also supporting the notion that methanogens play a primary role in the formation of MeHg in eutrophic lakes.

(3) Additionally, the higher contribution of methanogens to MeHg production under AOM input could also be supported by the results of microbial inhibition experiments (*i.e.*, experiment C). Notably, the amount of added BESA was insufficient for methanogen inhibition while the added Mo for sulfate reduction inhibition was in excess, partly due to the relatively high abundance of methanogens in eutrophic lake water[47,65] (discussed in Text S7). The inhibition experiments' results could thus be biased by underestimating the contribution of methanogens to MeHg production while overestimating that of SRB. Therefore, the observed comparable decreases in Hg methylation rate k_m in "Algae" (87.2% lower following methanogen inhibition and 81.2% lower following SRB inhibition, Figure S4) possibly suggest that methanogens (rather than SRB) could be more critical in Hg methylation under AOM input.

Our results suggest an important role for methanogens in Hg methylation in water from eutrophic shallow lakes, particularly during algal decomposition. The increased abundances and activities of methanogens in response to AOM input may be attributed to methanogens being more reactive to bioavailable carbon sources. The bioavailable fractions of AOM provide a series of ecological niches in which specialized populations could bloom.[66] The biotic complexity and the ever-changing redox conditions in lake water (especially during algal decomposition) generate a highly competitive environment;[47,67] In such a situation (*i.e.*, under AOM input), Archaea may occupy a far wider range of environmental niches than bacteria and exhibit better responsiveness, competitiveness, and utilization efficiency for bioavailable carbon sources,[68] leading to their key role in promoting microbial MeHg production. Recent studies showed that eutrophication could result in active methanogenesis, indicating that the enhanced abundance/activities of methanogens could be stimulated by carbon input associated with algal decomposition.[69,70] Similar to our observations, methanogens on periphyton in lake water have also been reported to act as major contributors to MeHg formation, possibly due to their better adaptation to the highly competitive environment created by periphytic biofilm in lakes.[47]

However, under oligotrophic conditions, the contributions of other microbial methylators (*e.g.*, SRB or FeRB) to Hg methylation cannot be ignored. The levels of MeHg under oligotrophic conditions (*e.g.*, the "Control" group with the Chl a content of 15.3 ± 1.76 $\mu g \cdot L^{-1}$) were positively correlated with the abundances of Archaea (R_{adj}^2 = 0.943, $p < 0.05$, Figure S9C) or Deltaproteobacteria (R_{adj}^2 = 0.911, $p < 0.05$, Figure S9D). This result indicates the importance of both Archaea (*e.g.*, methanogens) and Deltaproteobacteria (*e.g.*, SRB or FeRB) for MeHg production in water under oligotrophic conditions. Similarly, a recent study conducted in oligotrophic boreal lakes (the levels of Chl a were generally less than 10 $\mu g \cdot L^{-1}$) reported that *Methanomicrobiales* and *Geobacteraceae* were abundant members of Hg-methylating communities, and methanogens and FeRB contributed to more than half of the Hg methylation in boreal lakes.[71] Notably, researchers also pointed out that syntrophic metabolisms between methanogens and SRB became more important for Hg methylation (contributing approximately 72% of k_m) in the nutrient-poor peatlands.[23] Our results suggest that Archaea were more important than Deltaproteobacteria for MeHg production in the "Algae" treatments, while the results from the "Control" groups reveal that both Archaea and Deltaproteobacteria played important roles in MeHg production under oligotrophic conditions. Nevertheless, it should be noted the inconsistency may also partly come from the differences between the experimental conditions and the actual lake systems. It is necessary to further verify the specific contributions of Archaea and Deltaproteobacteria in the real lake environments.

4. ENVIRONMENTAL IMPLICATIONS

The intensity of algal blooms in lakes has increased globally over the past decades.[15] Especially in China, there are approximately 8700 km^2 of eutrophic lakes,[72] and most of them are distributed in eastern China.[28] Additionally, eastern China has a relatively higher Hg atmospheric deposition rates (*e.g.*, 50−100 $\mu g \cdot m^{-2} \cdot a^{-1}$) when compared to other regions.[31,73] Notably, atmospheric Hg can be methylated and rapidly accumulated in food chains (*e.g.*, 24 h after its deposition).[74] Our microcosm study, as well as *in situ* incubation, demonstrates that Hg methylation can be enhanced in water from eutrophic shallow lakes during algal decomposition. However, the model results indicate that the AOM-facilitated MeHg production by 1 or 2 orders of magnitude in water could be offset by AOM-induced MeHg biodilution, leading to moderate changes in MeHg bioaccumulation in fish (−12.2−11.7%) under eutrophication. Though the current health risk of exposure to MeHg by consumption of fish for residents in these eutrophic areas is considered low,[75] our study suggests that the risk could increase evidently under severe eutrophication conditions. Future study is needed to assess the potential changes in MeHg dietary exposure in these areas, which is of particular concern for people relying on fish as their primary protein source.

It should be noted that our most results were obtained from microcosm experiments. Although results from *in situ* incubation study were found to be consistent with laboratory observations, there are multiple factors that can regulate Hg methylation process in actual lake environments, including light penetration, wave disturbance, and ever-changing redox conditions.[18,47] On the other hand, in deep lakes, Hg methylation has been found to be significantly enhanced by settling planktonic particles in water column, for example, Lake Geneva (a maximum depth of 309 m) in Western Europe.[11] Therefore, attention to AOM-mediated Hg methylation should be paid to various aquatic systems such as deep lakes, coastal areas, or estuaries. Further studies are necessary to expand our results to the field, for example, the possible promotion of AOM-mediated Hg methylation in changing environments (*e.g.*, redox fluctuations), and the responses of AOM-induced microbial changes in substrates and metabolic types in the natural lake environments.

Remarkably, our results highlight that algal decomposition would lead to substantial changes in carbon supply, with potential impacts on microbial communities related to element cycling. Specifically, the distinctive fractions of AOM following algal decomposition can help shift a variety of ecological niches,[66] which would in turn impact microbial communities and microbial transformation of elements. For instance, in this study, we report that bioavailable fractions of AOM could facilitate the growth of Archaea and promote microbial MeHg

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production. In addition to Hg, the input of AOM has also been found to increase in arsenate reductase genes (*arr*A and *ars*C) and thus promote As(V) reduction to As(III).[76] Similarly, the decomposition of lake-dominated algae (*Chlorella pyrenoidosa* and *Microcystis aeruginosa*) provided abundant labile organic matter, which jointly elevated *mcrA* and the *mcrA*/*pmoA* ratio, enhancing CH_4 production.[69] More microbial researches, for example, by investigating short- and long-term microbial responses to various fractions of AOM as well as exploring their subsequent effects on element cycling, are needed in future studies.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.0c08395.

Measurements of THg and MeHg; fluorescence DOM measurements; DOM characterization by FTICR-MS; abundance analysis of microbial Hg methylators; preliminary experiment using specific inhibitors; determination of the Hg methylation rate; relative contributions of methanogens and SRB to Hg methylation; *in situ* incubation on Hg methylation; prediction of MeHg exposure risk by a model; detailed sampling information; treatments in the microcosm experiments; numbers and percentages of DOM fractions *via* FTICR-MS; molecular formulas for DOM *via* FTICR-MS; location of sampling sites from three eutrophic lakes; schematic diagram of the microcosm experiments; relative ratios of MeHg levels in water; effects of AOM and inhibitor addition on Hg methylation rates; dynamic changes of DOC concentrations; changes in the total intensity of DOM components; dynamic changes in the incremental abundance of *hgcA* gene; relationships between the Deltaproteobacteria-*hgcA* gene abundance and intensity of DOM components; relationships between *hgcA* gene abundance and the MeHg levels; dynamic activities of microbial Hg methylators; and relationships between CH_4 production and MeHg levels (PDF)

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Notes

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